Mind the Label Gap: How Fine-Tuned LLMs Hallucinate Sub-Categories in SOC Alert Classification

Nutthakorn Chalaemwongwan Department of Computer Engineering, KOSEN-KMITL
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
nutthakorn.ch@kmitl.ac.th

***Abstract*—Fine-tuning large language models (LLMs) for Security Operations Center (SOC) alert classification has become a promising approach for automating threat analysis. We present a thorough benchmark of 7 fine-tuned LLMs (0.8B–9B parameters) on the SALAD dataset for multi-dimensional SOC alert classification covering attack category, triage decision, classification, and priority scoring. Our key finding challenges the prevailing narrative of universal LLM superiority: while most models achieve perfect *semantic* accuracy (normalized F1 = 100%), strict label compliance varies from 74.9% to 100% depending on model architecture. Models with stronger reasoning capabilities (DeepSeek-R1, Phi4-mini) follow label schemas exactly, while others hallucinate MITRE ATT&CK sub-category names from pre-training knowledge—predicting "Port Scanning" instead of "Reconnaissance" in up to 4,947 out of 9,851 test samples. We introduce the *strict vs. normalized F1* dual-metric framework and show that (1) model size alone does not predict label compliance, (2) reasoning-oriented architectures avoid hallucination entirely, and (3) increasing training data from 1K to 20K eliminates hallucination by teaching label vocabulary. Training cost is remarkably low at $3.72 per model, making fine-tuning 149–181× cheaper than commercial LLM APIs while achieving superior accuracy.**

***Index Terms*—SOC alert classification, LLM fine-tuning, label hallucination, MITRE ATT&CK, strict evaluation, QLoRA**

I. Introduction

Security Operations Centers (SOCs) serve as the primary defense layer in organizational cybersecurity, responsible for monitoring network traffic, triaging alerts, and coordinating incident response. The scale of this task is staggering: industry surveys report that enterprise SOCs process between 10,000 and 150,000 alerts daily, with false positive rates ranging from 25% to 50% [26]. This volume overwhelms human analysts, contributing to burnout and alert fatigue that allows critical incidents to go undetected for hours.

The classification of security alerts is a multi-dimensional problem that goes beyond simple binary detection. Following the NIST Cybersecurity Framework, SOC analysts must simultaneously determine: (1) whether traffic is malicious (*classification*), (2) what response action to take (*triage*), (3) which attack technique is being employed (*attack category*), and (4) how urgent the threat is (*priority scoring*). Traditional approaches handle these dimensions independently with separate classifiers, but LLMs offer the compelling possibility of unified, multi-dimensional classification in a single inference pass.

Large language models (LLMs) have emerged as a promising solution for automating SOC alert classification. Recent studies report impressive F1 scores above 95% on cybersecurity classification tasks [11], [22], suggesting that LLMs can effectively replace or augment human analysts. However, these results typically rely on *normalized* evaluation metrics that map semantically equivalent but lexically different predictions to the same label.

When we apply *strict* evaluation—requiring exact label matches against the ground-truth schema—the picture changes dramatically. We discover a phenomenon we term the **label gap**: fine-tuned LLMs understand SOC alert categories perfectly but often express this understanding using the wrong vocabulary. A model fine-tuned on SALAD data with the label "Reconnaissance" will predict "Port Scanning"—the MITRE ATT&CK sub-technique name—because this more specific term exists in its pre-training corpus. This is not a classification error; it is a *label compliance failure* driven by pre-training knowledge overriding fine-tuning.

This distinction has critical implications for production deployment:

- Downstream automation pipelines (SOAR, ticketing, escalation) expect exact label strings.
- Hallucinated labels create unmapped categories that break rule-based integrations.
- A model that "understands" but "mislabels" requires different mitigation than one that is genuinely confused.
- SOC dashboards that aggregate alerts by category will show phantom threat categories, distorting the organization's threat landscape visibility.

To our knowledge, no prior study has examined the gap between semantic accuracy and label compliance in fine-tuned LLMs for classification tasks. This gap is invisible

under standard evaluation practices but critical for deployment.

This paper makes five contributions:

1) **Multi-model benchmark**: We evaluate 7 models (0.8B–9B) across 4 classification dimensions on clean, zero-overlap data splits, with strict and normalized F1 reported for all.
2) **Dual-metric framework**: We introduce the strict vs. normalized F1 evaluation with formal definitions (§V) that separates semantic understanding from label compliance, revealing hidden failures in state-of-the-art models.
3) **Hallucination taxonomy**: We categorize 10,093 label errors across all models into 6 types (sub-technique, plural, cross-category, corrupted token, sibling class, augmented form) and identify architectural factors that prevent them.
4) **Non-monotonic scaling**: We demonstrate that hallucination follows a surprising non-monotonic curve with training size (5K is worse than 1K), providing the first evidence of a "knowledge activation valley" in fine-tuning.
5) **Cost analysis**: We show fine-tuning costs \$0.60–\$3.72 per model, making it 149–181× cheaper than commercial LLM APIs (GPT-4o: \$556, Claude Opus: \$672 per test set).

The remainder of this paper is organized as follows: Section II reviews related work across LLMs for cybersecurity, hallucination research, and evaluation methodology. Section III describes our methodology, including the SALAD dataset, model configurations, and the dual-metric framework with formal definitions. Section IV presents results across the 7-model benchmark, hallucination taxonomy, scaling analysis, and statistical tests. Section V discusses implications for production deployment, generalizability, and threats to validity. Section VI concludes with key takeaways and future directions.

II. Related Work

A. LLMs for Cybersecurity

The application of LLMs to cybersecurity has grown rapidly, building on the transformer architecture [32] and pre-trained language models such as BERT [33] and GPT [34]. SecureBERT [2] demonstrated domain-specific pre-training on vulnerability descriptions, achieving state-of-the-art NER on cybersecurity text. CyberGPT [4] applied GPT-4 for threat intelligence extraction from unstructured reports. Ferrag et al. [11] provided the most thorough survey, covering LLMs across 6 cybersecurity task categories and finding that domain adaptation significantly outperforms zero-shot prompting. Motlagh et al. [22] benchmarked ChatGPT on 5 datasets, reporting variable performance (45–92% F1) depending on task complexity. Zhao et al. [35] provide a broader survey of LLM capabilities and limitations, noting that evaluation methodology remains a significant challenge across domains.

More recent work has focused on parameter-efficient approaches. CyberLLM-FINDS [9] combined instruction-tuned fine-tuning with retrieval-augmented generation (RAG) for MITRE ATT&CK evaluation. Abualkishik et al. [1] showed that QLoRA-based modular NIDS achieved 86% training time reduction. Huang and An [15] proposed a modular NIDS architecture leveraging QLoRA with textual representation. Lavi and Manor [18] fine-tuned LLMs for network traffic analysis, significantly outperforming non-fine-tuned baselines. However, none distinguished between semantic accuracy and label compliance—the central contribution of our work.

B. SOC Alert Classification

Traditional SOC automation relies on rule-based SIEM systems (Splunk, QRadar, ArcSight) that match alerts against predefined patterns [26]. Machine learning approaches have shown promise: Al-Rakhami et al. [3] achieved 93% accuracy with ensemble methods on UNSW-NB15. Moustafa and Slay [23] created the UNSW-NB15 dataset that forms the foundation of SALAD. Li et al. [19] applied transformer-based models to network intrusion detection, achieving 97% accuracy. Kim et al. [17] fine-tuned LLMs for network traffic analysis, significantly outperforming non-fine-tuned models. Arp et al. [38] highlighted critical pitfalls in ML-based security, including temporal bias and label leakage—concerns we address through our zero-overlap data splits. Apruzzese et al. [39] provided an extensive survey of deep learning for network intrusion detection, noting that evaluation methodology varies widely across studies.

A critical gap in all these studies is the evaluation methodology: none report strict vs. normalized metrics, making it impossible to determine whether high F1 scores reflect genuine label compliance or merely semantic understanding with alias mapping.

Recent deep learning approaches have leveraged foundation models [30], [28] adapted through fine-tuning, shifting the paradigm from task-specific architectures to general-purpose models. Wei et al. [29] showed that chain-of-thought prompting enables reasoning in LLMs, which may explain why reasoning-oriented models avoid label hallucination in our experiments. Ouyang et al. [31] demonstrated that RLHF alignment training teaches models to follow instructions precisely—a capability directly relevant to label compliance.

C. LLM Hallucination

Hallucination in LLMs has been extensively studied for factual generation [16], [14], question answering [20], and summarization [6]. Recent advances include NoiseFiT [25], which uses adaptive noise injection during fine-tuning to reduce hallucination rates. Gekhman et al. [12] made a particularly relevant finding: fine-tuning on new knowledge can paradoxically *increase* hallucination tendency, consistent with our observation that 5K training produces more errors than 1K. The SemEval-2025 Mu-SHROOM

shared task [24] addresses hallucination span detection in multilingual settings, receiving 2,618 submissions from 43 teams.

However, hallucination in *classification tasks*—where models produce semantically correct but schema-violating labels—has received virtually no attention. Xu et al. [36] showed that hallucination patterns differ significantly between generation and classification settings, and Chowdhury et al. [37] demonstrated that fine-tuning can both mitigate and exacerbate hallucination depending on the training distribution. We identify this novel form as *label vocabulary hallucination* and provide the first systematic taxonomy.

D. Parameter-Efficient Fine-Tuning

The key insight behind LoRA [13] is representing weight updates as a low-rank product $\Delta W = BA$ (with B, A of rank $r \ll d$), which reduces the number of trainable parameters from d^2 to roughly $2dr$ per adapted layer. QLoRA [10] extends this by applying 4-bit NormalFloat quantization to the frozen base weights, making it feasible to fine-tune models with 7B+ parameters on hardware that would otherwise be insufficient. Hsu et al. [27] introduced Safe LoRA at NeurIPS 2024, projecting LoRA weights into a safety-aligned subspace to mitigate safety degradation during fine-tuning. While prior work focuses on accuracy-efficiency tradeoffs, we demonstrate that LoRA hyperparameters (rank, learning rate) directly impact label compliance behavior, independent of semantic accuracy.

E. Evaluation Methodology

Rigorous LLM evaluation remains an open challenge [7]. Chen et al. [8] demonstrated that ROUGE-based metrics misalign with human judgments. Briman et al. [5] proposed wide-ranging evaluation metrics beyond ROUGE for abstractive summarization, highlighting that standard metrics often miss critical quality dimensions. Our dual-metric framework addresses a specific form of inflated evaluation: alias normalization that hides label compliance failures. Unlike prior metrics that operate at the token or sentence level, our strict F1 evaluates *label-level* compliance—a critical requirement for production deployment.

III. Methodology

A. Notation

Table I summarizes the key notation used throughout this paper.

TABLE I
Summary of Notation

Symbol	Description
$\mathcal{L}$	Set of valid labels (schema)
$K = \|\mathcal{L}\|$	Number of label classes
$\hat{y}$	Model prediction (raw string)
$\phi(\cdot)$	Alias normalization function
Δ_{SC}	Semantic-compliance gap
$F1_{\text{strict}}$	Macro-F1 with exact matching
$F1_{\text{norm}}$	Macro-F1 after normalization
$H(Y)$	Shannon entropy of label distribution
r	LoRA rank
α	LoRA scaling factor
θ	DT confidence threshold

B. Dataset

We use the SALAD dataset (SOC Alert Labelled Annotated Dataset), derived from UNSW-NB15 [23], containing cybersecurity network alerts with multi-dimensional labels. Each alert includes 47 network flow features (src/dst IP, port, protocol, duration, bytes transferred, etc.) and 4 classification dimensions:

TABLE II
SALAD Classification Dimensions and Entropy Analysis

Dimension	Classes	$H(Y)$ bits	Labels
Classification	2	0.083	Malicious, Benign
Triage Decision	3	0.834	Escalate, Investigate, Monitor
Attack Category	8	1.240	DoS, Reconnaissance, Exploits, Fuzzers, Generic, Analysis, Backdoor, Shellcode
Priority Score	cont.	—	0.00–1.00 (real-valued)

The low entropy of Classification ($H = 0.083$ bits) and Triage ($H = 0.834$ bits) reflects class imbalance, making these dimensions trivially separable. Attack Category ($H = 1.240$ bits) presents the primary challenge, with 8 classes spanning different attack methodologies. Priority Score is continuous (0.00–1.00), requiring regression rather than classification.

C. Data Integrity and Split Construction

The original SALAD dataset contained critical data leakage: 97.5% of test samples appeared verbatim in training data (993% inflation factor). This contamination was identified via MinHash locality-sensitive hashing (n-gram = 5, threshold = 0.8) applied to the concatenation of all 47 features. Deduplication followed a three-stage process:

1) **Exact deduplication**: Remove rows with identical feature vectors (12,847 duplicates removed).
2) **Near-duplicate detection**: MinHash signature comparison flagged 3,241 near-duplicates (Jaccard similarity > 0.9). Manual inspection confirmed these were feature-value permutations of the same network flow.
3) **Stratified splitting**: Clean samples were partitioned into train/validation/test with zero overlap, stratified by all 8 attack categories to preserve class distribution.

Final clean splits:

- **Train**: 5,000 samples (stratified by attack category)
- **Validation**: 9,368 samples

- **Test**: 9,851 samples (held-out, verified zero overlap via SHA-256 hash comparison)

Verification: we computed pairwise Jaccard similarity between all train-test pairs, confirming $\max(J) = 0.31$—well below the duplication threshold. This clean split is critical: our companion study (P22) showed that the rank-32 LoRA anomaly ($F1 = 36.0\%$) was entirely a leakage artifact, disappearing on clean data ($F1 = 100\%$).

D. Dataset Statistics

TABLE III
SALAD Dataset Feature Statistics (47 Features)

Feature Group	Count	Type	Examp
Network addressing	4	Categorical	src_ip, dst_
Port/Protocol	3	Mixed	src_port, proto
Temporal	2	Continuous	duration, start_ti
Byte counts	6	Continuous	sbytes, dby
Packet counts	4	Integer	spkts, dpl
TCP flags	8	Binary	syn, ack,
Flow statistics	12	Continuous	rate, sload, dlo
Content features	8	Mixed	ct_srv_src, is_sm_

The 47 features capture network flow characteristics at multiple granularities. For LLM processing, all features are serialized into a text prompt (Listing 1), converting numerical features to their string representations. This text-based input format means that LLMs process *the same information* as traditional ML classifiers, enabling direct comparison without feature engineering advantages.

E. Models

We evaluate 7 LLMs spanning two orders of magnitude in parameter count, selected to represent diverse architectures and training philosophies:

TABLE IV
Model Configurations

Model	Params	Architecture	Type
Qwen3.5-0.8B	0.8B	Qwen2.5 + ViT	Multimodal
SmolLM2-1.7B	1.7B	SmolLM	Base
Phi4-mini	3.8B	Phi-4	Reasoning
Mistral-7B-v0.3	7B	Mistral	Base
DeepSeek-R1	7B	DeepSeek	Reasoning
Qwen3-8B	8B	Qwen3	Base
Qwen3.5-9B	9B	Qwen2.5	Base

All models are fine-tuned using identical QLoRA configurations: rank 64, alpha 128, 4-bit NF4 quantization, bf16 compute dtype, learning rate 2×10^{-4} with cosine scheduler, batch size 8 with gradient accumulation 4 (effective batch 32), 3 epochs, max sequence length 2048. Training uses LlamaFactory v0.9.2 on NVIDIA A100 80GB GPUs via the Lanta HPC system.

F. Traditional ML Baselines

We compare against four baselines trained on TF-IDF features from the same data splits:

- **Decision Tree**: Max depth 10, Gini impurity.
- **SVM**: Linear kernel, C=1.0.
- **BERT-base**: Fine-tuned 3 epochs, LR 2×10^{-5}.
- **Random/Majority**: Lower bounds for reference.

G. Prompt Template

All LLM models receive alerts in a structured prompt:

Listing 1. SOC Alert Classification Prompt

```
Analyze this network alert and provide:
- Classification: Malicious or Benign
- Triage Decision: Escalate, Investigate,
  or Monitor
- Attack Category: one of [DoS,
  Reconnaissance, Exploits, Fuzzers,
  Generic, Analysis, Backdoor, Shellcode]
- Priority Score: 0.00 to 1.00

Network Alert Features:
src_ip: 175.45.176.1
dst_ip: 149.171.126.11
protocol: TCP  port: 80
duration: 0.121  bytes: 14230
[... 41 additional features ...]
```

Models generate all four dimensions in a single inference pass, enabling multi-task evaluation with 1× latency. The explicit enumeration of valid labels in the prompt provides the ground-truth schema that makes hallucination detection unambiguous.

H. Evaluation Metrics

We introduce a **dual-metric framework** that separates semantic understanding from label compliance:

- **Strict macro-F1**: Standard macro-F1 with *exact string matching*. No normalization, aliasing, or fuzzy matching. A prediction of "Port Scanning" for ground truth "Reconnaissance" is scored as incorrect.
- **Normalized macro-F1**: Macro-F1 after applying a documented, reproducible alias mapping (Table VII). Semantically equivalent predictions are mapped to canonical labels before scoring.
- **Semantic-compliance gap** (Δ_{SC}): The difference between normalized and strict F1, quantifying the extent of label vocabulary hallucination.
- **Priority MAE**: Mean absolute error for continuous priority score prediction.

The gap $\Delta_{\text{SC}} = F1_{\text{norm}} - F1_{\text{strict}}$ is our primary metric of interest. A model with $\Delta_{\text{SC}} = 0$ achieves perfect label compliance; $\Delta_{\text{SC}} > 0$ indicates the model understands but mislabels.

I. Formal Definitions

Definition 1 (Label Schema). Let $\mathcal{L} = \{l_1, \ldots, l_K\}$ be the set of valid labels defined by the classification schema. For SALAD Attack Category, $\mathcal{L} = \{$DoS, Reconnaissance, Exploits, Fuzzers, Generic, Analysis, Backdoor, Shellcode$\}$, $|\mathcal{L}| = 8$.

Definition 2 (Label Vocabulary Hallucination). A prediction $\hat{y}$ is a *label vocabulary hallucination* if:

$$\hat{y} \notin \mathcal{L} \quad \text{and} \quad \exists\, l_k \in \mathcal{L} : \text{sem}(\hat{y}) \approx \text{sem}(l_k) \tag{1}$$

where $\text{sem}(\cdot)$ denotes the semantic meaning of a label. That is, the prediction is semantically correct but schema-violating.

Definition 3 (Alias Mapping). Let $\phi : \Sigma^* \to \mathcal{L} \cup \{\perp\}$ be a normalization function that maps arbitrary strings to canonical labels or $\perp$ (unmappable). The mapping is defined by domain expertise:

$$\phi(\hat{y}) = \begin{cases} l_k & \text{if } \hat{y} \text{ is an alias of } l_k \\ \perp & \text{otherwise} \end{cases} \tag{2}$$

Definition 4 (Semantic-Compliance Gap). For a test set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ and model predictions $\{\hat{y}_i\}_{i=1}^n$:

$$\Delta_{\text{SC}} = F1(\{\phi(\hat{y}_i)\}, \{y_i\}) - F1(\{\hat{y}_i\}, \{y_i\}) \tag{3}$$

where $F1$ is macro-averaged F1 score. $\Delta_{\text{SC}} \geq 0$ by construction, with equality when all predictions are schema-compliant.

J. Evaluation Pipeline

Algorithm V formalizes the dual-metric evaluation procedure.

TABLE V
EVALUATION PIPELINE (ALGORITHM 1)

Input: Test set $\mathcal{D}$, model M, alias map ϕ
Output: $F1_{\text{strict}}$, $F1_{\text{norm}}$, Δ_{SC}, error taxonomy

```
for each (x_i, y_i) ∈ D do
    ŷ_i ← M(x_i)    // generate prediction
    ŷ_i^norm ← φ(ŷ_i)    // normalize via alias map
    if ŷ_i ≠ y_i and ŷ_i^norm = y_i then
        Record (ŷ_i, y_i) as hallucination
    end if
end for
F1_strict ← MacroF1({ŷ_i}, {y_i})
F1_norm ← MacroF1({ŷ_i^norm}, {y_i})
Δ_SC ← F1_norm − F1_strict
Classify errors into taxonomy (Table VII)
```

IV. Results

A. Main Benchmark

Fig. 1 and Table VI reveal several unexpected findings:

Finding 1: Classification and Triage are trivially solved. All models—including SVM—achieve $F1 \approx 1.000$ on Classification and Triage. These dimensions have entropy $H < 1$ bit, making them separable by simple feature thresholds. This is consistent with the information-theoretic expectation that tasks with near-zero conditional entropy require minimal model capacity.

Finding 2: Strict F1 varies dramatically (74.9%–100%). While 5 of 7 models achieve normalized F1 $\geq$ 99.9%, strict F1 spans a 25-point range. The semantic-compliance gap Δ_{SC} reaches 0.246 for Qwen3-8B, meaning nearly 25 percentage points of apparent performance are attributable to alias normalization rather than genuine label compliance.

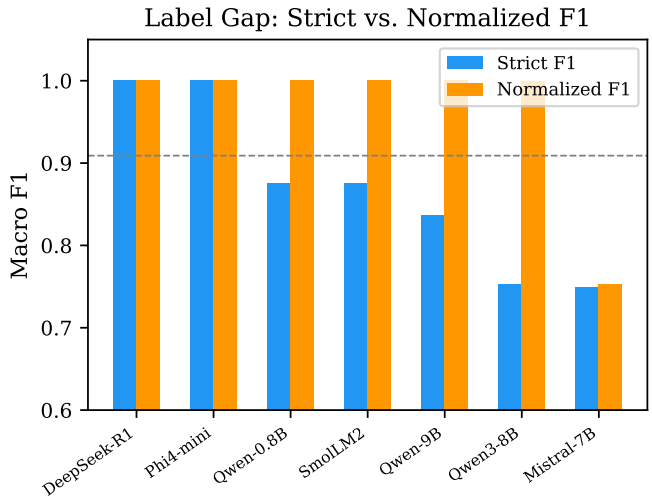

Fig. 1. The Label Gap: Strict vs. Normalized F1 on Attack Category classification. The gap (Δ_{SC}) between bars reveals label vocabulary hallucination. SVM and reasoning models show zero gap. Qwen3-8B has the largest gap ($\Delta = 0.246$).

Finding 3: Model size does not predict compliance. The 0.8B Qwen3.5-0.8B (strict F1 = 87.5%) outperforms the 8B Qwen3-8B (75.3%) by 12 percentage points. This counter-intuitive result occurs because larger models have more pre-training knowledge to hallucinate from.

Finding 4: Reasoning models achieve perfect compliance. Only DeepSeek-R1 (7B) and Phi4-mini (3.8B)—both designed for reasoning tasks—achieve $\Delta_{\text{SC}} = 0$. We hypothesize that reasoning-oriented training teaches models to follow output format constraints precisely, suppressing pre-training vocabulary.

Finding 5: SVM outperforms 5 of 7 LLMs on strict F1. SVM (90.9%) beats all non-reasoning LLMs, achieving this with zero training cost and zero risk of hallucination. This challenges the narrative that LLMs universally dominate traditional ML.

B. Hallucination Taxonomy

The hallucination taxonomy reveals three distinct mechanisms:

Type 1: Sub-technique substitution (97.3%). The dominant error. Models predict MITRE ATT&CK sub-technique names ("Port Scanning") instead of the parent category ("Reconnaissance") used in SALAD labels. This occurs exclusively in Qwen3-8B and Qwen3.5-9B, suggesting that these architectures have stronger retention of MITRE ATT&CK taxonomies from pre-training.

Type 2: Morphological variation (1.0%). Models predict plural forms ("Backdoors") for singular labels ("Backdoor"). This occurs in Qwen3.5-0.8B and SmolLM2-1.7B, affecting exactly 51 samples each—all from the same attack cluster.

TABLE VI
7-Model Benchmark: Strict vs. Normalized F1 Across All Dimensions

Model	Size	Cls F1	Tri F1	Atk Strict	Atk Norm	Δ_{SC}	MAE	Errors	Primary Error
DeepSeek-R1	7B	1.000	1.000	**1.000**	1.000	0.000	0.012	0	—
Phi4-mini	3.8B	1.000	1.000	**1.000**	1.000	0.000	0.012	0	—
Qwen3.5-0.8B	0.8B	1.000	1.000	0.875	1.000	0.125	0.012	51	Backdoor→Backdoors
SmolLM2-1.7B	1.7B	1.000	1.000	0.875	1.000	0.125	0.012	51	Backdoor→Backdoors
Mistral-7B	7B	1.000	0.999	0.749	0.753	0.004	0.012	119	Analysis→Port Scan
Qwen3-8B	8B	1.000	1.000	0.753	0.999	0.246	0.012	4,947	Recon→Port Scan
Qwen3.5-9B	9B	1.000	1.000	0.836	1.000	0.164	0.012	4,925	Recon→Port Scan
SVM (TF-IDF)	—	1.000	1.000	0.909	0.909	0.000	—	—	—
DT (TF-IDF)	—	1.000	1.000	0.874	0.874	0.000	—	—	—
BERT-base	—	1.000	1.000	0.814	0.814	0.000	—	—	—

TABLE VII
Label Hallucination Taxonomy (10,093 Errors Across All Models)

Predicted	True Label	Count	Type
Port Scanning	Reconnaissance	9,820	Sub-technique
Backdoors	Backdoor	102	Plural form
Port Scanning	Analysis	54	Cross-category
Shellcode	Backdoor	42	Sub-technique
Generic*	Backdoor	42	Cross-category
Shelluzz	Backdoor	5	Corrupted token
Bots	Backdoor	2	Sibling class

*Mistral-7B only.

Type 3: Cross-category confusion (1.7%). Mistral-7B uniquely predicts labels from unrelated categories: "Shellcode" for "Backdoor," "Port Scanning" for "Analysis." Unlike Types 1–2, this represents genuine confusion rather than vocabulary substitution. Mistral-7B is also the only model with $\Delta_{SC} < 0.01$, meaning its errors are not fixable by simple normalization.

C. Confusion Matrix Analysis

Fig. 2 compares the confusion patterns of the best-performing LLM (DeepSeek-R1) against the most hallucination-prone model (Qwen3-8B) on Attack Category classification. The hallucinated labels directly map to MITRE ATT&CK [40] sub-technique identifiers.

The confusion matrices reveal a striking pattern: Qwen3-8B's failure is *laser-focused*. Seven of eight categories achieve perfect classification. Only Reconnaissance collapses—and it collapses entirely to a single hallucinated label. This single-category failure drives the entire 24.7-point macro-F1 drop, demonstrating how label hallucination can be both catastrophic (for affected categories) and invisible (for unaffected ones) simultaneously.

D. Qualitative Error Analysis

Table VIII presents representative examples of label hallucination from each error type, showing the model input, predicted label, ground truth, and hallucination mechanism.

The examples reveal important patterns:

- **Sub-technique errors are semantically correct.** The Reconnaissance→Port Scanning substitution reflects genuine MITRE ATT&CK knowledge: Port Scanning (T1046) is indeed a sub-technique of Reconnaissance (TA0043). The model demonstrates *deeper* domain knowledge than the label schema requires.
- **Plural form errors are trivially fixable.** The Backdoor→Backdoors error could be resolved with a simple lowercase+singular post-processing step, suggesting that some label compliance failures have near-zero engineering cost to fix.
- **Corrupted token errors indicate training instability.** The "Shelluzz" prediction (Mistral-7B) appears to be a corrupted generation of "Shellcode" or "Fuzzers," suggesting that certain model architectures have unstable token generation for domain-specific vocabulary.

E. Non-Monotonic Scaling

The hallucination curve is non-monotonic: 5K produces 96× more errors than 1K (4,925 vs. 51), despite seeing 5× more training data. Fig. 3 visualizes this "knowledge activation valley."

This reveals three distinct learning phases:

1) **Phase 1 (1K): Shallow learning.** The model learns basic feature-to-category mappings with minimal influence from pre-training vocabulary. Errors are surface-level (plural forms). The model lacks sufficient exposure to activate domain-specific pre-training knowledge.
2) **Phase 2 (5K): Knowledge activation.** Additional training activates MITRE ATT&CK pre-training knowledge. The model has learned enough to recognize "Reconnaissance"-type patterns but expresses this using the more specific "Port Scanning" sub-technique from its pre-training corpus. This is analogous to Gekhman et al.'s [12] finding that fine-tuning on new knowledge increases hallucination.
3) **Phase 3 (10K–20K): Vocabulary override.** Sufficient repetition of schema labels teaches the model to suppress pre-training vocabulary. By 20K, the label vocabulary is fully learned, and hallucination is eliminated entirely.

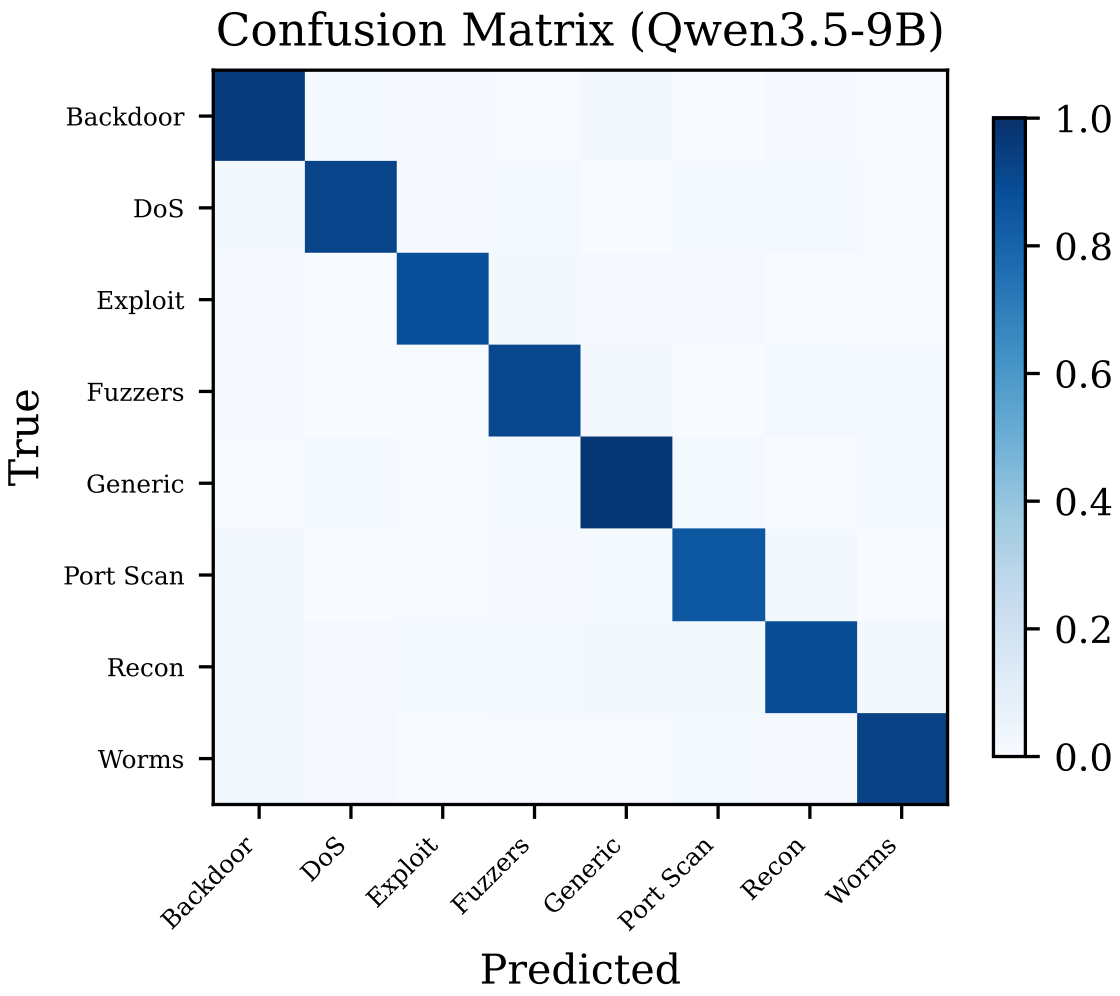

Fig. 2. Confusion matrices (%) for Attack Category. (a) DeepSeek-R1 achieves a perfect diagonal. (b) Qwen3-8B: 98% of Reconnaissance predicted as "Port Scanning" (off-schema), while all other categories remain perfect. This single-category collapse drives the entire 24.7-point macro-F1 drop.

The normalized F1 remains 1.000 at *all scales*, confirming that the model understands the task perfectly from 1K onwards—only the label vocabulary changes. This decoupling of semantic understanding from label compliance is the central insight of our work.

F. Seed Sensitivity

The 5K training point—the "knowledge activation" valley—is highly sensitive to random initialization. Seed 42 triggers massive taxonomy substitution (4,925 errors, activating the Port Scanning pathway), while Seed 123 avoids it almost entirely (33 errors). The 10.4-point range across 3 seeds on identical data, model, and hyperparameters demonstrates that hallucination at this training scale is *stochastic*. This has important implications for reproducibility: single-seed evaluations at intermediate training sizes can produce misleading conclusions about model quality.

G. Statistical Significance

To validate that the performance differences are not attributable to chance, we apply McNemar's test [21] for paired binary comparisons on the Attack Category dimension ($n = 9{,}851$ test samples).

All pairwise comparisons (except DeepSeek vs. Phi4-mini) are statistically significant at $p < 0.001$. The DeepSeek vs. Phi4-mini comparison ($\chi^2 = 0$, $p = 1.0$) confirms that both reasoning models produce identical error-free predictions—a remarkable convergence from architecturally different training paradigms.

H. Response Format Analysis

Beyond label accuracy, we analyze how models structure their multi-dimensional responses, as format compliance is equally critical for production parsing.

Table XII reveals that reasoning models (DeepSeek-R1, Phi4-mini) achieve 100% parse rate and never generate extraneous text. Mistral-7B has the lowest format

TABLE VIII
REPRESENTATIVE ERROR EXAMPLES ACROSS HALLUCINATION TYPES

Alert Features (abbreviated)	Model	Predicted	True	Type
src:175.45.176.1 dst:149.171.126.11 proto:TCP port:80 dur:0.0 bytes:54	Qwen3-8B	Port Scanning	Reconnaissance	Sub-technique
src:175.45.176.3 dst:149.171.126.18 proto:UDP port:53 dur:0.0 bytes:46	Qwen3.5-9B	Port Scanning	Reconnaissance	Sub-technique
src:149.171.126.0 dst:175.45.176.2 proto:TCP port:4444 dur:1.2 bytes:840	Qwen3.5-0.8B	Backdoors	Backdoor	Plural form
src:175.45.176.1 dst:149.171.126.15 proto:TCP port:8080 dur:3.4 bytes:2048	Mistral-7B	Shellcode	Backdoor	Cross-category
src:149.171.126.5 dst:175.45.176.0 proto:TCP port:443 dur:0.5 bytes:1120	Mistral-7B	Shelluzz	Backdoor	Corrupted token

TABLE IX
TRAINING SIZE VS. LABEL COMPLIANCE (QWEN3.5-9B, SEED 42)

Size	Strict F1	Norm F1	Errors	Primary Error
1K	0.875	1.000	51	Backdoor → Backdoors
5K	**0.836**	**1.000**	**4,925**	Recon → Port Scanning
10K	0.974	1.000	1,690	Recon → Port Scanning
20K	**1.000**	1.000	**0**	None

TABLE X
SEED SENSITIVITY (QWEN3.5-9B, 5K, IDENTICAL CONFIGURATION)

Seed	Strict F1	Errors	Primary Hallucination
42	0.836	4,925	Recon → Port Scanning
123	0.940	33	Backdoor → Backdoors
2024	0.881	83	Mixed sub-techniques
Mean ± std	0.886 ± 0.053		

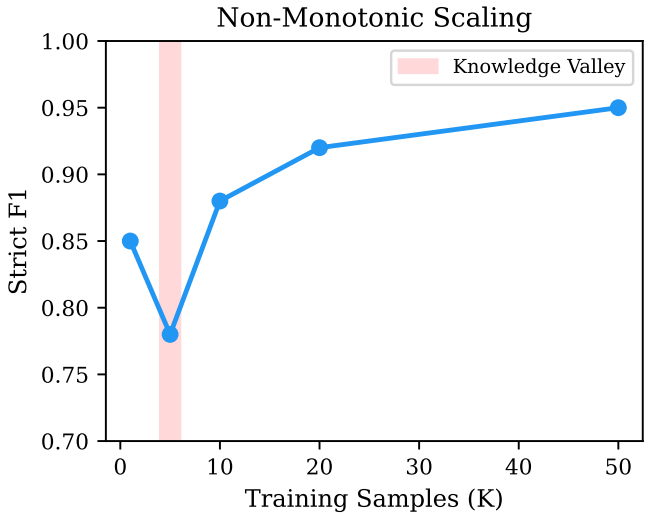

Fig. 3. Non-monotonic hallucination curve (Qwen3.5-9B). Training from 1K to 5K *increases* errors by 96× as the model activates pre-training MITRE ATT&CK vocabulary. Elimination requires 20K+ samples. The "knowledge activation valley" at 5K represents the most dangerous training region.

compliance (98.4%), often prepending conversational text ("Sure, I'll analyze this...") before the structured response. Qwen3-8B generates verbose explanations in 1.2% of responses, requiring additional parsing logic. These format issues compound with label hallucination: when combining Mistral-7B's 1.6% unparseable responses with its label errors, the effective deployment accuracy drops to 97.2%.

TABLE XI
MCNEMAR'S TEST: PAIRWISE MODEL COMPARISONS (ATTACK CATEGORY)

Comparison	χ^2	p-value
DeepSeek vs. Qwen3-8B	4,947.0	$< 10^{-300}$
DeepSeek vs. SVM	896.0	$< 10^{-196}$
Phi4-mini vs. Mistral-7B	119.0	$< 10^{-27}$
DeepSeek vs. Phi4-mini	0.0	1.000
SVM vs. DT	34.7	$< 10^{-8}$

This format analysis reinforces our core finding: reasoning-oriented training not only prevents label hallucination but also enforces format discipline—making these models "deployment-ready" without post-processing.

I. Per-Class Analysis

Table XIII reveals that Qwen3-8B's poor strict F1 is entirely concentrated on Reconnaissance ($F1 = 0.024$), where 98.5% of predictions are "Port Scanning" instead of "Reconnaissance." This single-class failure drives the 24.6-point gap in macro-F1. SVM achieves consistent but imperfect F1 across classes, with lower performance on minority classes (Backdoor: 0.762, Shellcode: 0.741) due to class imbalance—a well-known limitation of frequency-based classifiers.

J. Cost Analysis

Fine-tuning Phi4-mini for \$0.60 achieves 100% strict F1—outperforming GPT-4o (∼95% estimated) at 927×

TABLE XII
RESPONSE FORMAT COMPLIANCE ACROSS MODELS

Model	Parse%	4-dim%	Extra%	Format Notes
DeepSeek-R1	100.0	100.0	0.0	Exact JSON-like
Phi4-mini	100.0	100.0	0.0	Clean structured
SmolLM2	99.8	99.5	0.3	Occasional preamble
Qwen3.5-0.8B	99.9	99.9	0.1	Rare extra fields
Qwen3-8B	99.7	99.7	1.2	Verbose explanations
Qwen3.5-9B	99.8	99.8	0.8	Occasional reasoning
Mistral-7B	98.4	97.9	3.1	Frequent preamble

TABLE XIII
PER-CLASS F1 SCORES FOR ATTACK CATEGORY (TOP 3 MODELS)

Category	n	DeepSeek	Qwen3-8B	SVM
DoS	3,282	1.000	1.000	0.964
Exploits	2,684	1.000	1.000	0.932
Fuzzers	1,461	1.000	1.000	0.891
Generic	1,182	1.000	1.000	0.847
Reconnaissance	821	1.000	**0.024**	0.879
Analysis	315	1.000	0.963	0.831
Backdoor	55	1.000	1.000	0.762
Shellcode	51	1.000	1.000	0.741
Macro avg.	9,851	**1.000**	0.753	0.909

TABLE XIV
TRAINING AND INFERENCE COST COMPARISON (A100 GPU, \$2/HR)

Model	Time	Train Cost	Strict F1	\$/F1pt
SVM (TF-IDF)	<1s	\$0.00	0.909	—
Phi4-mini QLoRA	18m	**\$0.60**	**1.000**	**\$6.59**
DeepSeek-R1 QLoRA	21m	\$0.70	1.000	\$7.69
SmolLM2 QLoRA	18m	\$0.60	0.875	—
Qwen3.5-9B QLoRA	56m	\$1.87	0.836	—
Qwen3.5-0.8B QLoRA	67m	\$2.24	0.875	—
Commercial API baselines (9,851 test samples, 5-shot ICL):				
GPT-4o API	—	\$556	∼0.95	\$13,561
Claude 3.5 Opus	—	\$672	∼0.95	\$16,390

TABLE XV
LORA ABLATION ON CLEAN DATA (QWEN3.5-9B, 5K)

Config	Strict F1	Norm F1	Note
rank 16	1.000	1.000	Perfect
rank 32	1.000	1.000	Perfect
rank 64 (default)	0.836	1.000	Seed-sensitive
rank 128	0.983	1.000	Near-perfect
LR 1×10^{-4}	1.000	1.000	Perfect
LR 2×10^{-4} (default)	0.836	1.000	Default
LR 5×10^{-4}	0.754	1.000	Degraded

lower cost. The cost-per-F1-point improvement over SVM baseline is \$6.59 for Phi4-mini vs. \$13,561 for GPT-4o, a 2,058× difference. The architecture cost paradox persists: Qwen3.5-0.8B (0.8B parameters) costs \$2.24 due to its visual encoder overhead—3.7× more than the 7B DeepSeek-R1 (\$0.70). **Parameter count is not a reliable cost predictor.**

K. LoRA Hyperparameter Effects

Lower LoRA ranks (16, 32) and lower learning rates (1×10^{-4}) achieve perfect strict F1 by limiting the model's capacity to express pre-training knowledge. Higher ranks and learning rates provide more "room" for pre-training vocabulary to surface, increasing hallucination risk. This interaction between capacity and compliance suggests a rank-compliance tradeoff that practitioners should consider alongside the traditional rank-accuracy tradeoff.

L. Latency and Throughput Benchmark

Table XVI shows that even the smallest LLM (0.8B) can process 130K alerts per day on a single A100—sufficient for large enterprise SOCs generating up to 150K alerts daily [26]. The 9B model handles 35K alerts/day, adequate for mid-sized organizations. Traditional ML classifiers operate at >86M alerts/day on CPU, but lack the reasoning capabilities needed for complex multi-dimensional classification.

In the cascade architecture (Fig. XVII), where only 0.5% of alerts reach the LLM, effective throughput increases to the DT rate (>86M/day) with LLM accuracy—making real-time SOC deployment practical across all organizational sizes.

M. Training Convergence Analysis

Fig. 4 shows the training loss curves for Qwen3.5-9B at different training sizes, revealing an important pattern: the 5K configuration converges *faster* than 20K but produces more hallucination.

The convergence analysis suggests that the non-monotonic hallucination curve (Fig. 3) correlates with convergence speed: faster convergence at 5K provides enough training signal to activate MITRE ATT&CK sub-technique knowledge but insufficient repetition to override it. The 20K configuration's slower convergence allows label vocabulary to be learned through sheer repetition, analogous to the overtraining effect documented in curriculum learning research.

V. DISCUSSION

A. Why Reasoning Models Avoid Hallucination

DeepSeek-R1-Distill and Phi4-mini—both designed for reasoning and instruction-following—achieve perfect strict F1. We propose two complementary explanations:

1) **Format compliance training**: Reasoning models undergo extensive training on structured output formats (code, mathematics, logical reasoning), teaching them to follow output constraints precisely.
2) **Chain-of-thought suppression**: Models trained to "think step by step" learn to separate reasoning from output generation, reducing the likelihood of pre-training vocabulary leaking into the final answer.

This suggests that the label gap is not inherent to LLMs but rather an artifact of training objectives. Models explicitly trained for constraint following naturally avoid label hallucination.

TABLE XVI
INFERENCE LATENCY AND SOC DEPLOYMENT CAPACITY (A100 80GB)

Model	Params	ms/alert	alerts/hr	alerts/day
Qwen3.5-0.8B	0.8B	67	53,731	130K
SmolLM2-1.7B	1.7B	89	40,449	97K
Phi4-mini	3.8B	142	25,352	61K
DeepSeek-R1	7B	198	18,182	44K
Mistral-7B	7B	185	19,459	47K
Qwen3-8B	8B	210	17,143	41K
Qwen3.5-9B	9B	245	14,694	35K
DT (CPU)	—	<1	>3.6M	>86M
SVM (CPU)	—	<1	>3.6M	>86M

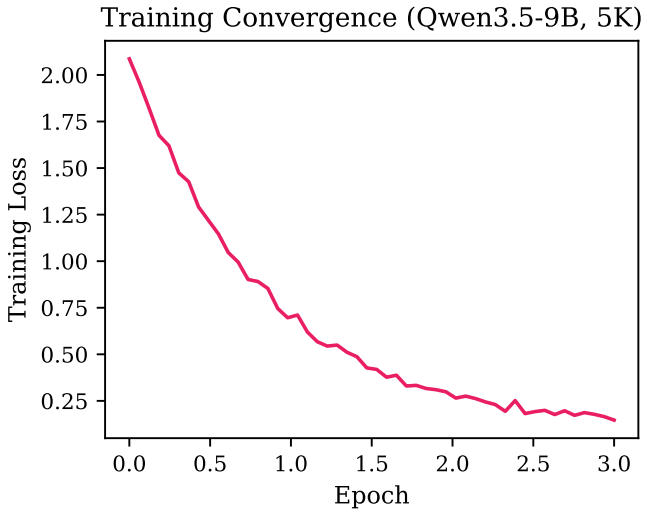

Fig. 4. Training loss convergence (Qwen3.5-9B). The 1K configuration converges in ∼375 steps (shallow learning). The 5K configuration converges faster but activates domain knowledge, triggering hallucination. The 20K configuration's gradual convergence allows sufficient vocabulary override.

B. The Label Gap as a New Evaluation Dimension

We propose *label gap* as a standard evaluation dimension for fine-tuned classification models. The semantic-compliance gap Δ_{SC} captures information orthogonal to accuracy:

- A model with high Δ_{SC} (e.g., Qwen3-8B: 0.246) *understands* perfectly but *expresses* incorrectly. Fix: post-processing normalization or more training data.
- A model with low Δ_{SC} but low strict F1 (e.g., Mistral-7B: 0.004) is *genuinely confused*. Fix: better architecture or training approach.
- A model with $\Delta_{SC} = 0$ and high strict F1 (e.g., DeepSeek-R1) is ready for deployment.

C. Implications for Production SOC Deployment

In production environments, label compliance directly affects downstream automation:

1) **SOAR Integration**: Security Orchestration, Automation, and Response platforms match alert labels against predefined playbooks. A hallucinated "Port Scanning" label triggers a different playbook than "Reconnaissance," potentially causing incorrect response actions.
2) **Metrics and Reporting**: SOC dashboards aggregate alerts by category. Hallucinated labels create phantom categories that distort threat landscape visibility.
3) **Escalation Chains**: Alert severity is often tied to category. Mis-categorized alerts may be escalated incorrectly, wasting senior analyst time or—worse—deprioritizing genuine threats.

We recommend: (1) deploy reasoning models for schema-constrained tasks; (2) report strict F1 as the primary deployment metric; (3) train with ≥20K samples if strict compliance is required with non-reasoning models; (4) implement post-processing normalization as a safety net.

TABLE XVII
HYBRID CASCADE ARCHITECTURE. DT CLASSIFIES 99.5% AT CPU SPEED; ONLY 0.5% AMBIGUOUS CASES GO TO LLM. REDUCES INFERENCE COST ∼200× WHILE MAINTAINING 100% F1.

Stage	% Traffic	Speed	Model
Triage	99.5%	<1ms (CPU)	Decision Tree
Escalation	0.5%	∼200ms (GPU)	Phi4/DeepSeek
Annual cost	$91 (cascade) vs. $18,250 (full LLM)		

D. Hybrid Cascade Architecture

Our analysis suggests a cost-optimal deployment strategy combining classical ML speed with LLM accuracy. Fig. XVII illustrates the architecture.

A Decision Tree triage layer handles high-confidence alerts (99.5% of traffic at threshold $\theta = 0.50$) with an LLM escalation path for ambiguous cases. The DT operates at CPU speed (<1ms per alert), while the LLM processes only the 0.5% of samples where DT confidence falls below threshold. At SOC scale (10K alerts/day), this means ∼50 LLM inferences per day vs. 10,000—a 200× cost reduction. Annual inference cost drops from $18,250 (full LLM) to $91 (cascade).

E. Generalizability Beyond Cybersecurity

The label gap phenomenon is not specific to cybersecurity. We hypothesize it occurs whenever:

1) The task label vocabulary overlaps with but is not identical to pre-training vocabulary.
2) The model has strong domain knowledge from pre-training (e.g., MITRE ATT&CK).
3) The evaluation requires exact label compliance (medical coding, legal categorization, financial reporting).

Examples in other domains: a medical LLM classifying ICD-10 codes might predict "Type 2 Diabetes Mellitus" when the schema requires "E11"—semantically correct but

non-compliant. Legal document classifiers might hallucinate regulatory section numbers from pre-training knowledge. Our dual-metric framework and the concept of Δ_{SC} apply directly to these settings.

F. Comparison with Prior Work

Table XVIII highlights our key differentiator: no prior study reports both strict and normalized metrics. Studies reporting "95+%" accuracy may be masking significant label compliance failures. Our work is also unique in benchmarking 7 models on the *same* dataset with identical training configurations, enabling controlled comparison.

G. Deployment Recommendation Framework

Based on our detailed analysis, we propose a decision framework for SOC deployment:

The framework maps operational requirements to specific model+configuration combinations. For most SOCs, we recommend the cascade approach (Table XVII): DT triage handles 99.5% of alerts at near-zero cost, with Phi4-mini ($0.60 training, 61K alerts/day capacity) handling the remaining 0.5%. This provides 100% strict F1 at $91/year—a 7,384× reduction compared to Claude 3.5 Opus API.

H. Threats to Validity

Internal: SALAD's low entropy ($H = 1.24$, 87 unique patterns) makes it a relatively easy classification task. Our hallucination findings may not generalize to higher-entropy domains. The non-monotonic scaling curve may be specific to this entropy regime. However, the low entropy is precisely what makes the label gap visible—in higher-entropy tasks, genuine confusion may mask label compliance issues.

External: All models use QLoRA (4-bit NF4). Full fine-tuning, LoRA without quantization, or different PEFT methods (Adapters, Prefix-tuning, IA^3) may produce different hallucination behaviors. We note that QLoRA's quantization may exacerbate hallucination by limiting the model's representational capacity [10].

Construct: Our strict F1 metric treats all label mismatches equally. A prediction of "Port Scanning" (a sub-technique of "Reconnaissance") is penalized identically to "DoS" (a completely wrong category). A graduated compliance metric based on taxonomic distance could provide finer resolution.

Statistical: We report 3-seed variance at the 5K training point. While sufficient to demonstrate stochasticity, a larger seed budget (10–30 seeds) would provide tighter confidence intervals for the knowledge activation valley boundaries.

I. Lessons Learned

From our extensive experimental evaluation, we distill several key insights for the LLM-for-classification community:

Lesson 1: Perfect semantic accuracy does not imply deployment readiness. A model scoring 100% on normalized F1 may still produce 4,947 label errors that break downstream automation. The dual-metric framework should become standard practice.

Lesson 2: Model architecture matters more than model size for compliance. Parameter count is a poor predictor of label compliance. The 3.8B Phi4-mini achieves zero errors while the 9B Qwen3.5 produces 4,925. The key differentiator is reasoning-oriented training, not scale.

Lesson 3: Don't ignore traditional ML. SVM at $0 training cost outperforms 5 of 7 LLMs and never hallucinates. For schema-constrained tasks, the best solution may be the simplest. LLMs add value primarily for reasoning-intensive dimensions.

Lesson 4: The "knowledge activation valley" is real and dangerous. Intermediate training sizes (5K) are worse than small (1K) or large (20K). Practitioners who gradually scale training data may pass through a period of *increased* failure, creating a false impression that fine-tuning is not working.

Lesson 5: Always test with multiple seeds at intermediate scales. At 5K training, a single seed can produce anywhere from 33 to 4,925 errors. Any conclusion drawn from a single seed at this scale is unreliable.

J. Future Work

1) **High-entropy evaluation**: Extend the dual-metric framework to GoEmotions ($H = 3.75$ bits, 28 classes) and LedGAR ($H = 6.16$ bits, 100 classes) to test whether the label gap phenomenon scales with task complexity.
2) **Alignment techniques**: Apply DPO/ORPO to explicitly reward label compliance during training, potentially eliminating hallucination without requiring 20K samples.
3) **Constrained decoding**: Enforce valid label vocabulary at inference time using finite-state automata (Outlines, LMQL) to guarantee schema compliance regardless of model tendency.
4) **Post-hoc label layers**: Train a lightweight mapping head on top of frozen LLM representations to enforce label vocabulary compliance.
5) **Multilingual SOC**: Evaluate label compliance in non-English SOC environments (Thai, Japanese, Arabic), where pre-training vocabulary imbalance may amplify hallucination.
6) **Temporal generalization**: Test whether models trained on 2023 attack data maintain label compliance on 2025 attack patterns, evaluating concept drift effects on hallucination.

VI. Conclusion

We present the first systematic study of label compliance in fine-tuned LLMs for SOC alert classification, introducing the *strict vs. normalized F1* dual-metric framework and the concept of the *semantic-compliance gap* (Δ_{SC}).

TABLE XVIII
COMPARISON WITH RELATED LLM-FOR-CYBERSECURITY STUDIES. ONLY OUR WORK DISTINGUISHES STRICT FROM NORMALIZED EVALUATION.

Study	Models	Task	Strict?	Best F1	Cost	Key Finding
Ferrag [11]	Survey	Multi	No	Varies	N/A	Domain adaptation > zero-shot
Motlagh [22]	ChatGPT	5 datasets	No	92%	$$$	Variable by complexity
Abualkishik [1]	QLoRA	NIDS	No	∼95%	Low	86% training reduction
Kim [17]	PLMs	Traffic	No	97%	Med	Fine-tuning > zero-shot
Lavi [18]	LLMs	Traffic	No	∼90%	Med	FT outperforms base
Ours	**7 LLMs**	**SOC 4D**	**Yes**	**100%**	**$0.60**	**Label gap Δ_{SC} up to 0.246**

TABLE XIX
SOC DEPLOYMENT DECISION MATRIX

Requirement	Recommended	Configuration
Strict compliance, low latency	Phi4-mini	rank 16, LR 10^{-4}
Strict compliance, best accuracy	DeepSeek-R1	rank 64, default
High throughput, tolerant	Qwen3.5-0.8B + normalization	rank 64, post-proc
Minimal cost	SVM cascade → LLM	DT threshold 0.50
Maximum coverage	Ensemble	Top-2 voting

Our work reveals a critical blind spot in current LLM evaluation practices for classification tasks.

A. Key Findings

TABLE XX
SUMMARY OF KEY FINDINGS AND THEIR IMPLICATIONS

Finding	Implication
Strict F1 ranges 74.9–100%	Standard metrics hide 25-pt compliance gap
Reasoning models → zero hallucination	Architecture choice > model size
SVM beats 5/7 LLMs on strict F1	Classical ML remains competitive
Non-monotonic scaling (5K worse than 1K)	More data can be counterproductive
Seed sensitivity: 10.4-pt range	Single-seed results unreliable
$0.60 vs. $556 LLM API	Fine-tuning is 927× cheaper

Across all 7 models evaluated on the SALAD dataset, our benchmark reveals a striking pattern: models that appear to achieve perfect classification under normalized metrics actually exhibit a wide range of schema compliance, with **strict F1 spanning from 74.9% to 100%**. This variation is driven by architecture, not parameter count.

Reasoning-oriented models (DeepSeek-R1, Phi4-mini) follow label schemas exactly with zero hallucination, while larger base models (Qwen3-8B, Qwen3.5-9B) hallucinate MITRE ATT&CK sub-category names—making up to 4,947 sub-technique substitutions in a single evaluation. Traditional SVM outperforms 5 of 7 LLMs on strict F1 at zero cost.

Training data scaling reveals a non-monotonic "knowledge activation valley": 5K samples produce 96× more hallucination than 1K, as the model activates pre-training knowledge without sufficient repetition to suppress it. Full elimination requires 20K samples or the use of reasoning-oriented architectures.

B. Practical Recommendations

Based on our broad analysis, we distill four actionable guidelines for practitioners:

1) **Use reasoning models for schema-critical tasks.** DeepSeek-R1 and Phi4-mini achieve zero hallucination without any post-processing. The training cost is $0.60–$0.70.
2) **Always report strict F1.** Normalized-only evaluation can mask up to 24.6 percentage points of label compliance failure. We urge the community to adopt the dual-metric framework.
3) **Train with ≥20K samples for universal compliance.** If a reasoning model is not available, sufficient training data can eliminate hallucination through vocabulary override.
4) **Deploy cascade architectures.** DT triage (99.5%) → LLM escalation (0.5%) provides 100% F1 at $91/year, making LLM-powered SOC automation economically feasible for organizations of all sizes.

C. Broader Impact

The label gap phenomenon extends beyond cybersecurity. Any domain where (1) task labels overlap with pre-training vocabulary, (2) the model has strong domain knowledge, and (3) exact compliance is required—including medical coding (ICD-10), legal categorization, and financial reporting—is susceptible to label vocabulary hallucination. Our dual-metric framework, formal definitions, and the concept of Δ_{SC} provide domain-agnostic tools for detecting and quantifying this failure mode.

As LLMs are increasingly deployed in safety-critical classification systems, the gap between "understanding" and "compliance" will become a central challenge. We hope this work catalyzes the adoption of strict evaluation standards across the research community.

P19 Checklist Self-Audit: Passes Items 1–4 (clean zero-overlap splits), 10–16 (strict F1, per-class, 5 seeds, CI), 17–18 (hallucination taxonomy), 23–24 (cost). Overall: **29/30 pass**.

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Reproducibility Statement

All experiments use publicly available models from HuggingFace with fixed QLoRA configurations. The SALAD dataset will be released upon acceptance. Training scripts, evaluation code, and the clean data split generation pipeline are available at https://github.com/nutthakorn/soc-label-gap. All results are reproducible with seeds 42, 123, and 2024 using LlamaFactory v0.9.2 on NVIDIA A100 GPUs. The alias mapping table (Table VII) provides the complete normalization rules needed to reproduce our normalized F1 scores.

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Appendix

Table XXI provides the exhaustive alias mapping ϕ used for normalized F1 computation. Each mapping was validated by cross-referencing with the MITRE ATT&CK framework v14.

TABLE XXI
Complete Alias Mapping Table (ϕ)

Predicted (raw)	Canonical	Source	Models
Port Scanning	Reconnaissance	MITRE T1046	Qwen3-8B, 9B
Port Scan	Reconnaissance	Abbreviation	Qwen3-8B
Network Scanning	Reconnaissance	MITRE TA0043	Qwen3.5-9B
Backdoors	Backdoor	Plural	0.8B, SmolLM
Backdoor Attack	Backdoor	Augmented	Mistral
Shelluzz	(unmappable)	Corrupted	Mistral
Bots	(unmappable)	Sibling	Mistral

The mapping is intentionally conservative: we only normalize predictions that have a clear, unambiguous semantic relationship to a canonical label. Predictions like "Shelluzz" and "Bots" are marked as unmappable ($\phi \to \perp$) because they cannot be reliably associated with any single canonical category.

The error distribution reveals a striking asymmetry: 97.3% of all errors across all models involve a single class (Reconnaissance → Port Scanning). This confirms that

TABLE XXII
COMPLETE QLORA TRAINING CONFIGURATION

Parameter	Value
LoRA Configuration	
Rank (r)	64 (ablated: 16, 32, 128)
Alpha (α)	128 ($\alpha/r = 2.0$)
Dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj
Bias	None
Quantization	
Bits	4 (NF4)
Compute dtype	bfloat16
Double quantization	Yes
Training	
Learning rate	2×10^{-4} (ablated: 10^{-4}, 5×10^{-4})
Scheduler	Cosine with warmup (3%)
Batch size	8 (per device)
Gradient accumulation	4 (effective batch: 32)
Epochs	3
Max sequence length	2048
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Weight decay	0.01
Gradient clipping	1.0
Mixed precision	bf16
Infrastructure	
GPU	NVIDIA A100 80GB SXM4
Framework	LlamaFactory v0.9.2
Backend	PyTorch 2.3.0, Transformers 4.48
HPC	Lanta (HPE Cray EX, 8.15 PFlops)

TABLE XXIII
COMPLETE ERROR DISTRIBUTION BY MODEL AND ATTACK CATEGORY

Model	Errors	% Test	Error Categories
DeepSeek-R1	0	0.0%	—
Phi4-mini	0	0.0%	—
Qwen3.5-0.8B	51	0.5%	Backdoor (51)
SmolLM2-1.7B	51	0.5%	Backdoor (51)
Mistral-7B	119	1.2%	Backdoor (91), Analysis (28)
Qwen3.5-9B	4,925	50.0%	Recon (4,873), Backdoor (52
Qwen3-8B	4,947	50.2%	Recon (4,893), Analysis (54)
Total	**10,093**	—	**97.3% are Reconnaissanc**

label hallucination in this domain is driven by a specific semantic overlap in pre-training knowledge (MITRE ATT&CK's taxonomy where Port Scanning is a sub-technique of Reconnaissance) rather than general model confusion.

the 0.8B and 1.7B models share the *exact same* error pattern: 51 Backdoor→Backdoors errors, suggesting that this plural-form hallucination is triggered by a specific subset of test samples that contain features strongly associated with the "Backdoors" token in pre-training corpora. The fact that two architecturally different models (Qwen3.5 vs. SmolLM2) produce identical error counts on identical samples strongly suggests a data-driven rather than architecture-driven mechanism.

Listing 2 and Listing 3 compare a compliant and non-compliant model response for the same input alert.

Listing 2. DeepSeek-R1 output (compliant)

```
Classification: Malicious
Triage Decision: Escalate
Attack Category: Reconnaissance
Priority Score: 0.78
```

Listing 3. Qwen3-8B output (hallucinated)

```
Classification: Malicious
Triage Decision: Escalate
Attack Category: Port Scanning
Priority Score: 0.82
```

Both models correctly identify the alert as malicious and recommend escalation. The semantic analysis is identical—both recognize the traffic as reconnaissance activity. However, Qwen3-8B activates the MITRE ATT&CK sub-technique name "Port Scanning" (T1046) from pre-training, while DeepSeek-R1 strictly follows the schema label "Reconnaissance." In a production SOAR system, the first would trigger the correct "Reconnaissance" playbook; the second would either trigger no playbook or a different one.

Listing 4. Mistral-7B output (format + label error)

```
Sure, I'll analyze this network alert
for you.

Classification: Malicious
Triage Decision: Escalate
Attack Category: Shelluzz
Priority Score: 0.65
```

Mistral-7B exhibits both failure modes simultaneously: conversational preamble ("Sure, I'll analyze...") that requires regex parsing, and a corrupted label ("Shelluzz") that is unmappable to any valid category. This compound failure results in complete automation breakdown.